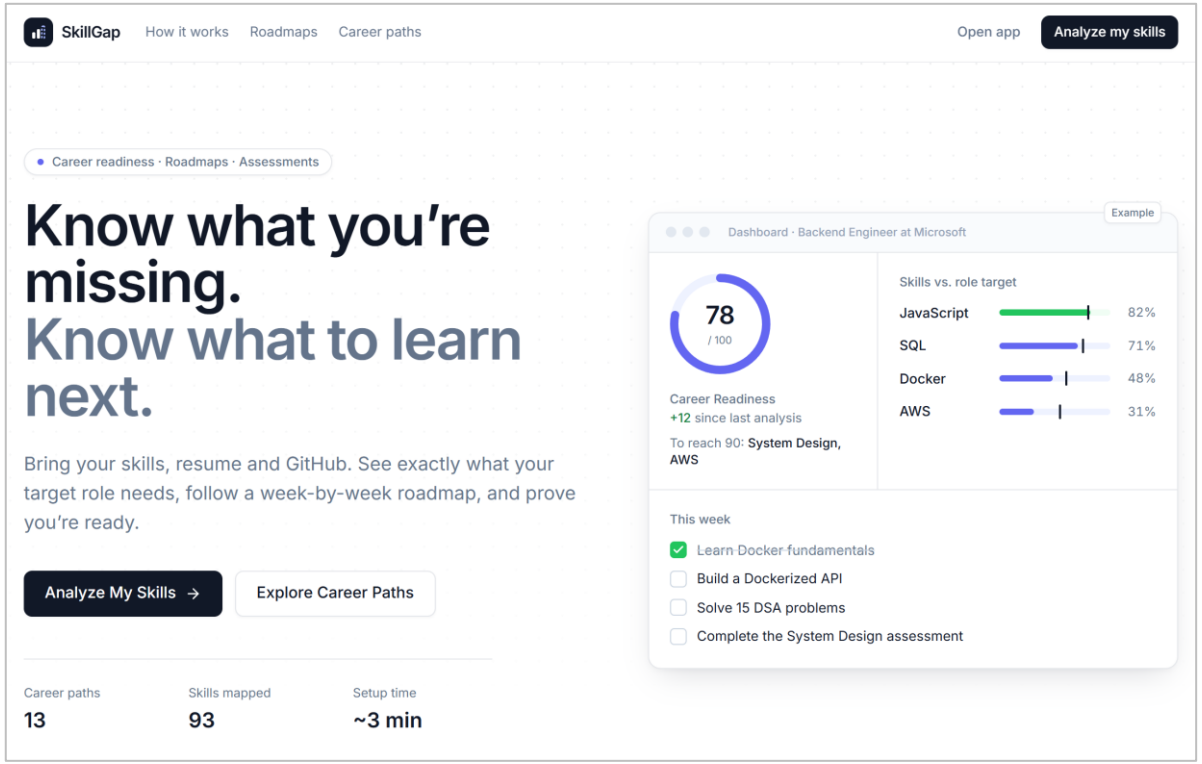
SkillGap Analyzer

“Know what you’re missing.
Know what to learn next.”

A web app that compares your skills with what a target job role expects, ranks the gaps, and turns them into a week-by-week learning roadmap.

13 career paths · 93 skills · 26 skill assessments

Live: skillgapanalyzer-alpha.vercel.app



Homepage of the live site

[Team name] · [Member names] · [Institution]

The problem: knowing what to learn next

Most students know the job title they want, but not the path to it.

The question students face	How SkillGap Analyzer answers it
1. What does the role actually need? Job posts list many tools at once and rarely say what level is expected.	13 career paths, each listing its required skills and a target level from 0 to 100
2. Where do I stand against it? A resume lists skills, but not which ones are below the expected level, or by how much.	Skill Gap page: your level next to the target for every requirement, ranked by priority
3. What should I learn first? Some skills depend on others and study time is limited, so learning becomes random.	Roadmap: prerequisites first, sized to the hours you have each week

Our solution

One guided flow takes a student from a target role to a weekly study plan.

1. Choose a role
13 roles or a custom one, plus an optional target company

→

2. Add your skills
Pick and self-rate them, or import from a resume or GitHub

→

3. Analyse the gap
Readiness score out of 100, gaps ranked by priority

→

4. Get a roadmap
Week-by-week phases in prerequisite order

→

5. Prove progress
Assessments update skill levels and re-plan the roadmap

Without a plan	With SkillGap Analyzer
Guess which skills a role needs	Each role lists its required skills and target levels
Learn topics in any order	Prerequisites always come before the skills that need them
No way to tell if you are improving	Readiness score changes after assessments and finished work
Plans ignore the time you have	Plan is sized to your daily hours and warns if it will not fit the deadline

Who it is for

Who uses it

College students
preparing for campus placements

Fresh graduates
aiming for a first job role

Career switchers
moving to a new track, e.g. Data Analyst to Data Engineer

Onboarding asks for your current level (Beginner, Intermediate or Advanced), so the pacing changes for each group.

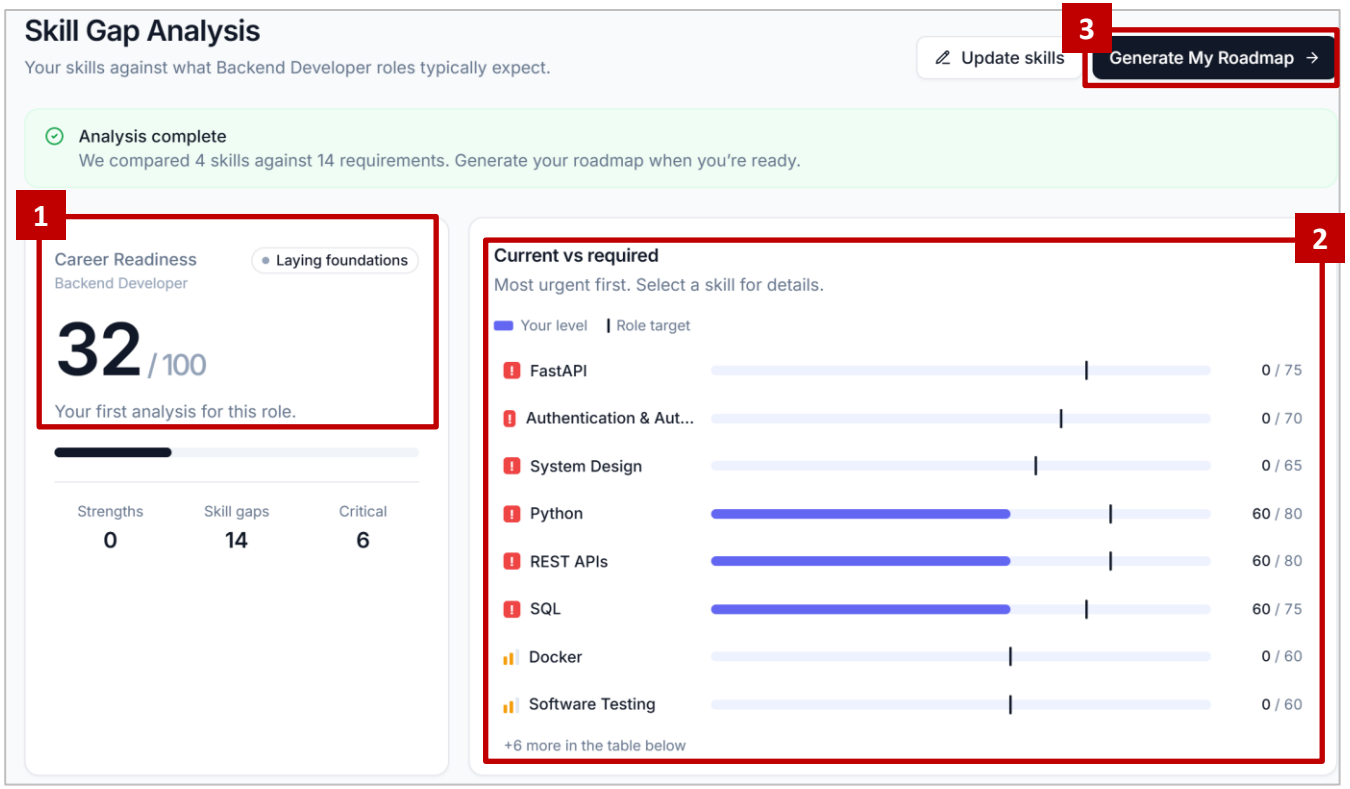
Three use cases

Situation	What they do on the site	What they get
Preparing for a role before placements	Pick Backend Developer, add 4 skills, choose 1 hour a day for 3 months	Readiness 32/100, 6 critical gaps and a 9-phase roadmap*
Applying to one specific job	Jobs → Job match: paste the job post and click Analyze job	A match % and a gap list for that job; “Target this job” plans for it
Choosing between roles or companies	Jobs → Career simulator	Match and preparation time for each role–company combination

* From our own test run on the live site, October 2026.

Key features on the live website

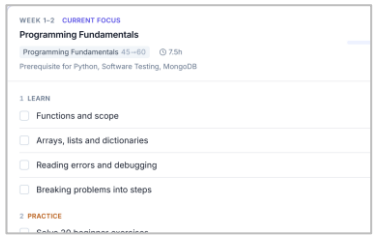
Skill gap analysis (the main screen after onboarding)



1 Readiness score out of 100 and its band

2 Your level (bar) against the role target (tick) for each skill

3 One click turns the ranked gaps into a roadmap



Dependency-aware roadmap

Weekly phases; each skill runs Learn → Practice → Build → Assess.

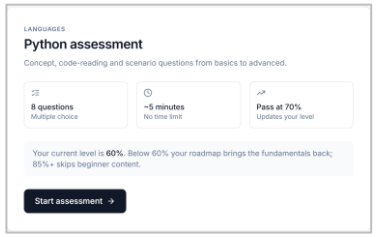
Skill Gap → Generate My Roadmap



Job description match

Paste a real job post to get a match % and a gap list for that job.

Jobs → Job match



Skill assessments

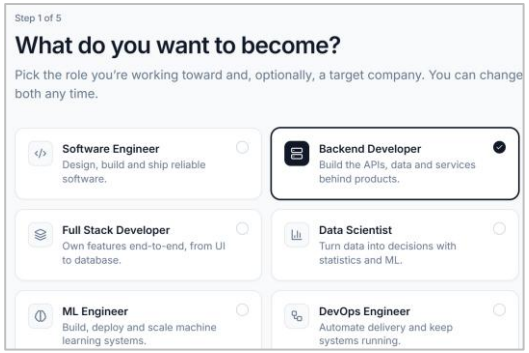
26 auto-graded tests of 8 questions; the score updates your level.

Assessments page

Also live: resume analyzer, GitHub analyzer, interview prep and a career simulator.

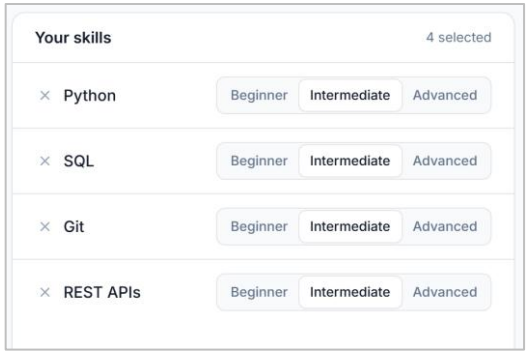
How the website works

The real flow on the live site. There is no sign-up: your data is saved in your browser.



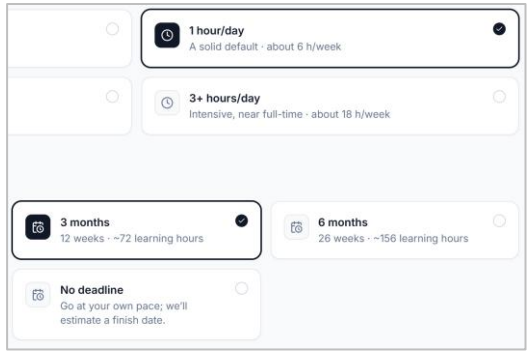
1. Pick a target role

Onboarding step 1: one of 13 roles or a custom role; company optional.



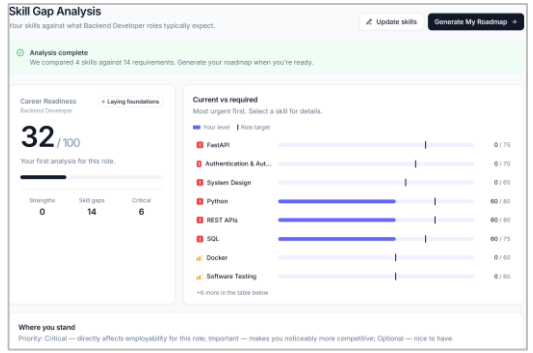
2. Add and rate skills

Step 3: rate each skill Beginner, Intermediate or Advanced, or import a resume.



3. Set your study time

Step 5: hours per day and a timeline, then Analyze My Skill Gap.



4. Review the gaps

Readiness score and ranked gaps. Next: Generate My Roadmap.

5. Follow the plan. The Roadmap page holds every phase; the Dashboard shows this week's tasks, sized to your weekly hours.

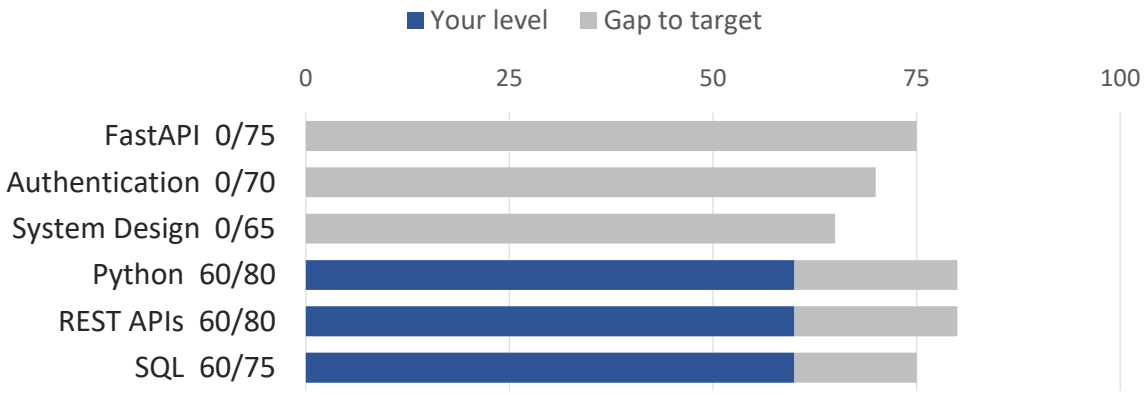
Not shown: step 2 asks your current level; step 4 reads an optional public GitHub profile.

How the gap and the roadmap are calculated

Measuring the gap

- Self-rating → level 0–100: Beginner 30, Intermediate 60, Advanced 85
- Each role sets a target and a weight: Critical 4, High 3, Medium 2, Low 1
- Priority (Critical, Important, Optional) comes from weight and gap size

Your level vs the role target, from our test run (Backend Developer)



32 / 100

readiness in the test run

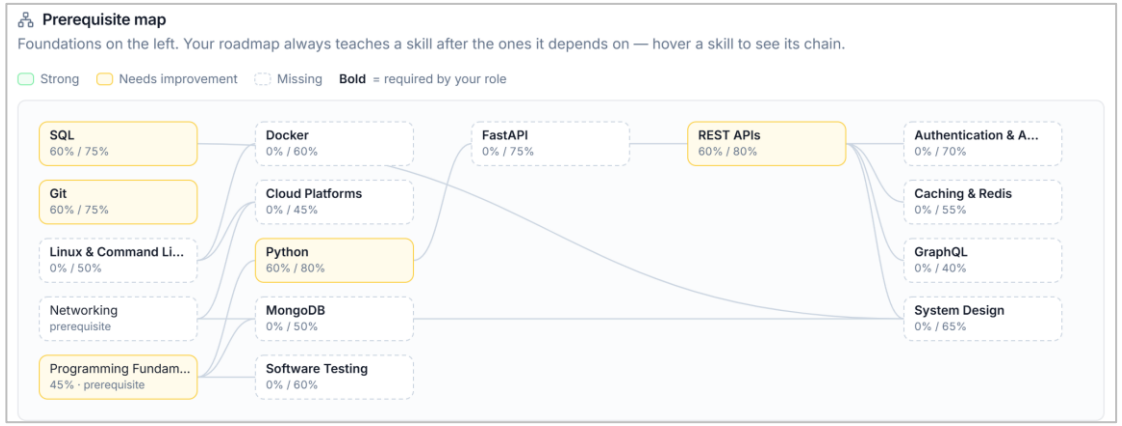
Readiness = the weighted share of the role's targets you already cover.

Σ weight × min(1, level ÷ target) ÷ Σ weight × 100

Building the roadmap

Ranked gaps → Add missing prerequisites → Order by dependency → Size hours to the gap → Fit to weekly time

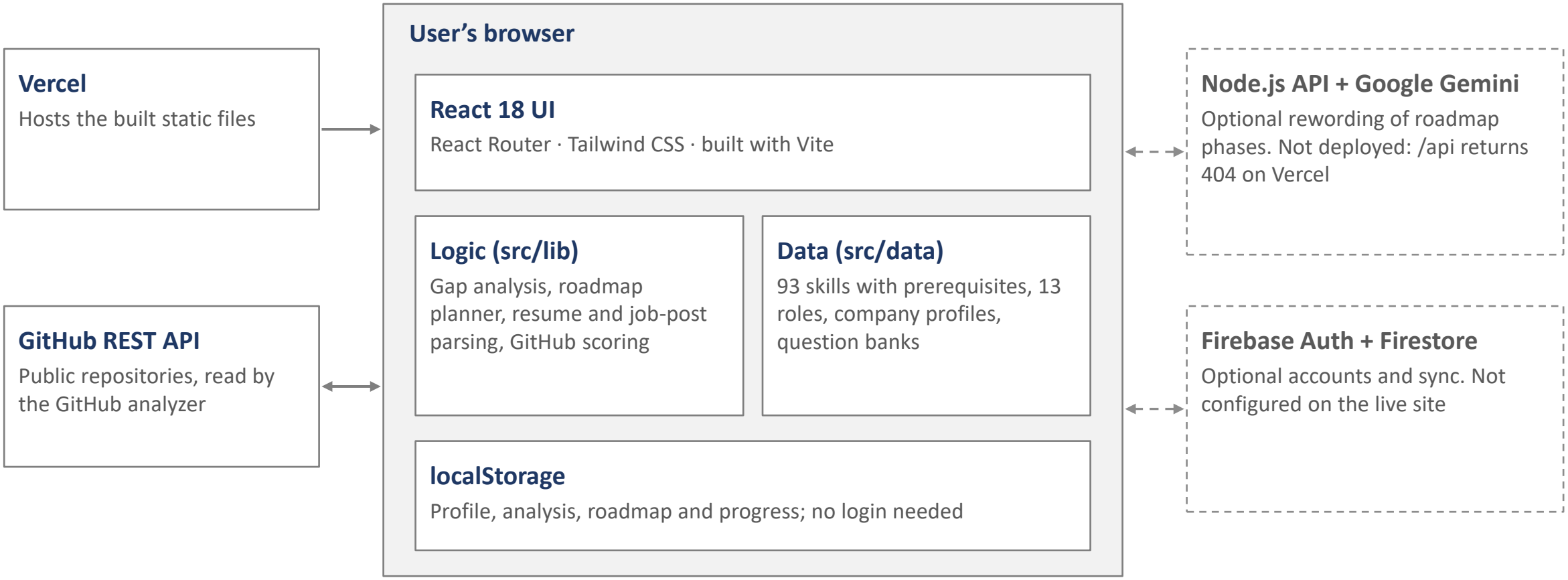
Each skill then runs **Learn → Practice → Build → Assess**



Result in our test run

- 9 roadmap phases, about 125.5 hours, roughly 21 weeks at 6 h/week
- Longer than the 12-week target, so the app warned about it and deferred 6 lower-priority skills

Technology stack and architecture



Solid boxes are running on the live site. Dashed boxes are in the code but not enabled on the live site.

All the analysis and planning are plain functions with no randomness, so they run in the browser without a server and always give the same result for the same input.

Impact, limitations and future scope

Expected impact

Intended benefits, not yet measured

- A vague goal becomes a ranked list of specific gaps
- A clear order to learn in, sized to the time available
- A quick fit check against a real job post before applying
- Assessments turn self-ratings into evidence

Current limitations

Found in the code and on the live site

- Levels start as self-ratings until an assessment is taken
- Role targets and company profiles are set by our team, not taken from live job data
- Data stays in one browser; no accounts on the live site
- Opening an inner page URL directly (e.g. /dashboard) shows a 404
- Resume reader needs text-based PDFs; GitHub checks are rate-limited

Future scope

Proposed, not built or not deployed yet

- Turn on accounts and sync (Firebase support is already in the code)
- Deploy the API so the optional Gemini step works live
- Add a Vercel rewrite rule so every page URL loads
- Check role targets with placement-cell or recruiter input, then add more roles

Conclusion and live demo

The problem

Students know the role they want, but not what they are missing or what to learn first.

What we built

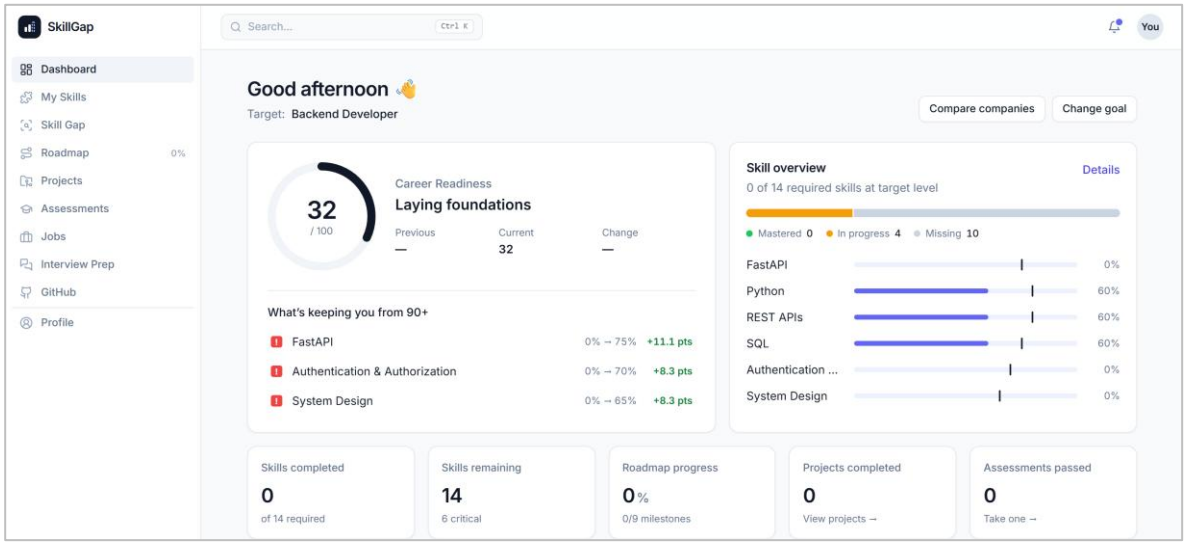
A working web app that measures skills against 13 roles, ranks the gaps and builds a week-by-week plan that respects prerequisites and available time.

Main takeaway

Every recommendation can be explained: it comes from the role's targets and a rule-based planner.

Live demo: skillgapanalyzer-alpha.vercel.app

Thank you. Questions?



Dashboard on the live site after our test run

In the demo

1. Pick a role and add a few skills
2. See the readiness score and ranked gaps
3. Generate the week-by-week roadmap

[Team name] · [Member names] · [Institution]